

References

- Childress-Thompson, R. (2016). A framework for assessing the reusability of hardware (reusable rocket engines) (NASA/TP-2017-00606). National Aeronautics and Space Administration.
<https://ntrs.nasa.gov/api/citations/20170000606/downloads/20170000606.pdf>
- Cook, S. (1995, January). The reusable launch vehicle technology program. In AIAA Aerospace Planes and Hypersonics Technologies Conference (AIAA Paper 95-6153). NASA Marshall Space Flight Center.
<https://ntrs.nasa.gov/api/citations/19950058868/downloads/19950058868.pdf>
- De Oliveira, A., & Lavagna, M. (2024). Coupling of advanced guidance and robust control for the descent and precise landing of reusable launchers. *Aerospace*, 11(11), 914.
<https://doi.org/10.3390/aerospace11110914>
- Dominguez Calabuig, G. J., Wilson, A., Bi, S., Vasile, M., Sippel, M., & Tajmar, M. (2024). Environmental life cycle assessment of reusable launch vehicle fleets: Large climate impact driven by rocket exhaust emissions. *Journal of Environmental Management*, 362, 120899. <https://doi.org/10.1016/j.jenvman.2024.120899>
- Dorney, D. (2022). Launch vehicle reusability in NASA crewed spaceflight [Presentation]. NASA Technical Reports Server.

https://ntrs.nasa.gov/api/citations/20220002635/downloads/NASA_LV_Reuse_final_1.pdf

Dresia, K., Jentzsch, S., Waxenegger-Wilfing, G., Hahn, R. D. S., Deeken, J., Oschwald, M., & Mota, F. (2021). Multidisciplinary design optimization of reusable launch vehicles for different propellants and objectives. *AIAA Journal*. <https://doi.org/10.2514/1.A34944>

Gstattenbauer, G. J. (2006). Cost comparison of expendable, hybrid, and fully reusable launch vehicles [Master's thesis, Air Force Institute of Technology].

Jo, B.-U., & Ahn, J. (2022). Optimal staging of reusable launch vehicles for minimum life cycle cost. *Aerospace Science and Technology*, 125, 107703. <https://doi.org/10.1016/j.ast.2022.107703>

Kang, S.-Y., Jo, M.-S., Choi, J.-Y., & Yang, S. S. (2025). Cost effectiveness of reusable launch vehicles depending on the payload capacity. *Aerospace*, 12, 364. <https://doi.org/10.3390/aerospace12050364>

Koelle, D. E., & Janovsky, R. (2007, September 10–13). Development and transportation costs of space launch systems [Conference paper]. 1st CEAS European Air and Space Conference, Ottobrunn, Germany.

Leser, P., Warner, J., Parker, P., Cramer, E., & Taylor, D. (2025, June 16). Probabilistic fracture analysis challenges and best practices [Presentation]. NASA Technical Reports Server.
https://ntrs.nasa.gov/api/citations/20250006182/downloads/2025-06-16_reusable_engine_TIM.pdf

Li, K., Zhang, W., Yuan, H., Zhou, J., & Ma, Y. (2025). Mission re-planning of reusable launch vehicles under throttling fault in the recovery flight based on controllable set analysis and a deep neural network. *Aerospace*, 12(3), 166.
<https://doi.org/10.3390/aerospace12030166>

National Aeronautics and Space Administration. (2024). Fracture control requirements for spaceflight hardware (NASA-STD-5019A w/ Change 4).
<https://standards.nasa.gov/sites/default/files/standards/NASA/w/CHANGE-4/4/NASA-STD-5019A-w-Change-4-mod-7-29-2024.pdf>

Russell, R., Park, A., Wells, D., West, B., McEnerney, B., Glendening, A., Lanigan, E., & Tilson, W. (2022, March 3). NASA's certification and qualification strategies for additively manufactured hardware: Reuse and challenges [Presentation]. NASA Technical Reports Server.
https://ntrs.nasa.gov/api/citations/20220002973/downloads/NASA_Russell_Aerospace_1.pdf

Ryan, R. G., Silver, B. J., Maloney, C. S., Wilson, C., & Marais, E. A. (2022). Impact of rocket launch and space debris air pollutant emissions on stratospheric ozone and climate.

Earth's Future, 10(7), e2021EF002612. <https://doi.org/10.1029/2021EF002612>

Sagliano, M., Fari, S., Máces Hernández, J., Marchetti, F., Harms, H., Heidecker, A., Woicke, S., Redondo Gutierrez, J., Seelbinder, D., & Dumont, E. (2025). Operational atmospheric landing guidance for reusable rockets [Conference paper]. AIAA SciTech 2025 Forum. <https://doi.org/10.2514/6.2025-2597>

Sethi, S. (2025, February 12). Personal interview.

Stappert, S., Simioana, M., & Sippel, M. (2020, October 19). Technical report on different RLV return modes' performances (Deliverable D2.1, Rev. 2). German Aerospace Center (DLR).

Webb, R. (2016). Is it worth it? The economics of reusable space transportation (NASA Technical Report). NASA. <https://ntrs.nasa.gov/api/citations/20160013369/downloads/20160013369.pdf>

Wertz, J. R. (2000, October 2–6). Economic model of reusable vs. expendable launch vehicles [Conference paper]. IAF Congress, Rio de Janeiro, Brazil.

Zapata, E. (2017). An assessment of cost improvements in the NASA COTS/CRS program and implications for future NASA acquisition and launch cost. NASA Technical Report (20170008895).

<https://ntrs.nasa.gov/api/citations/20170008895/downloads/20170008895.pdf>